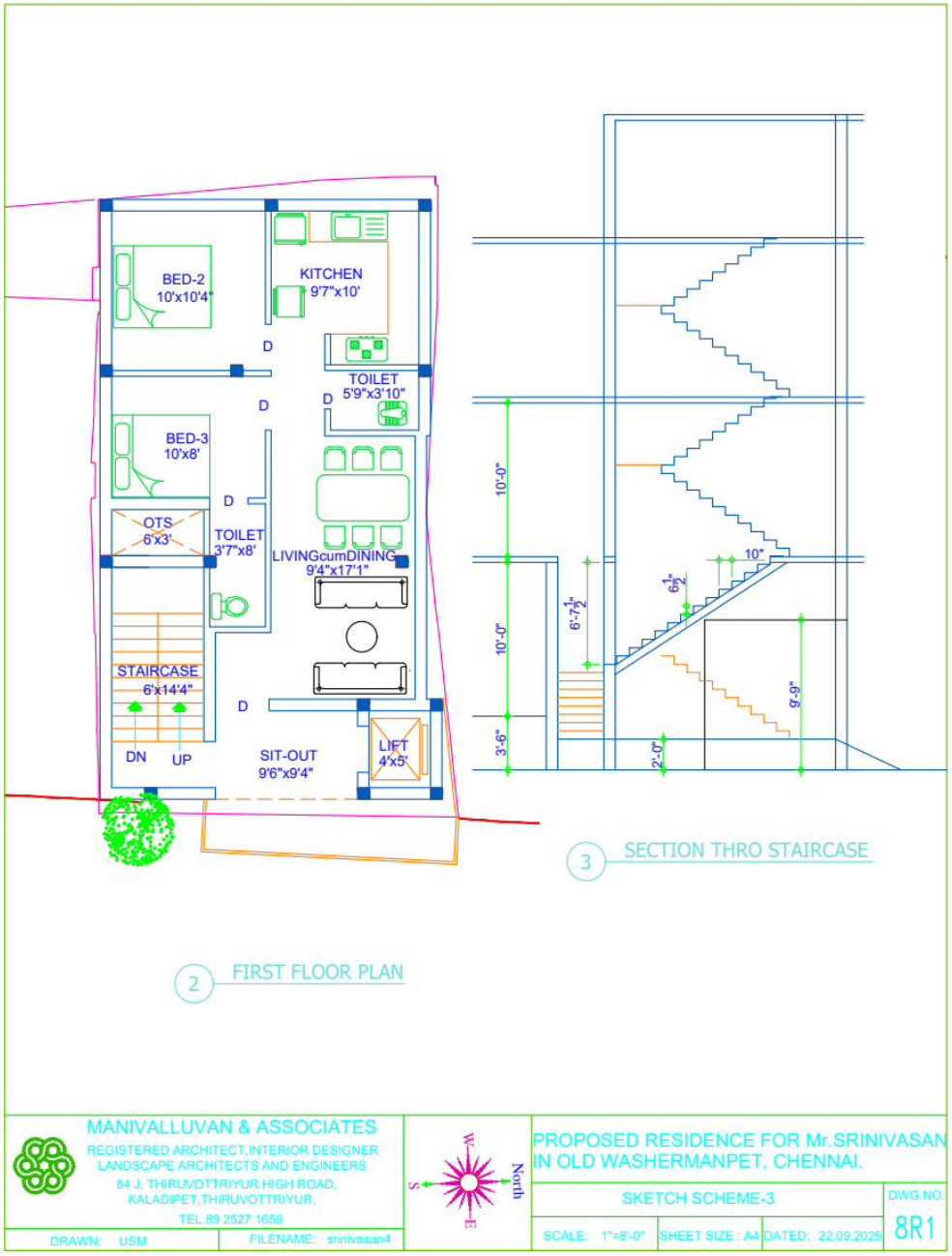
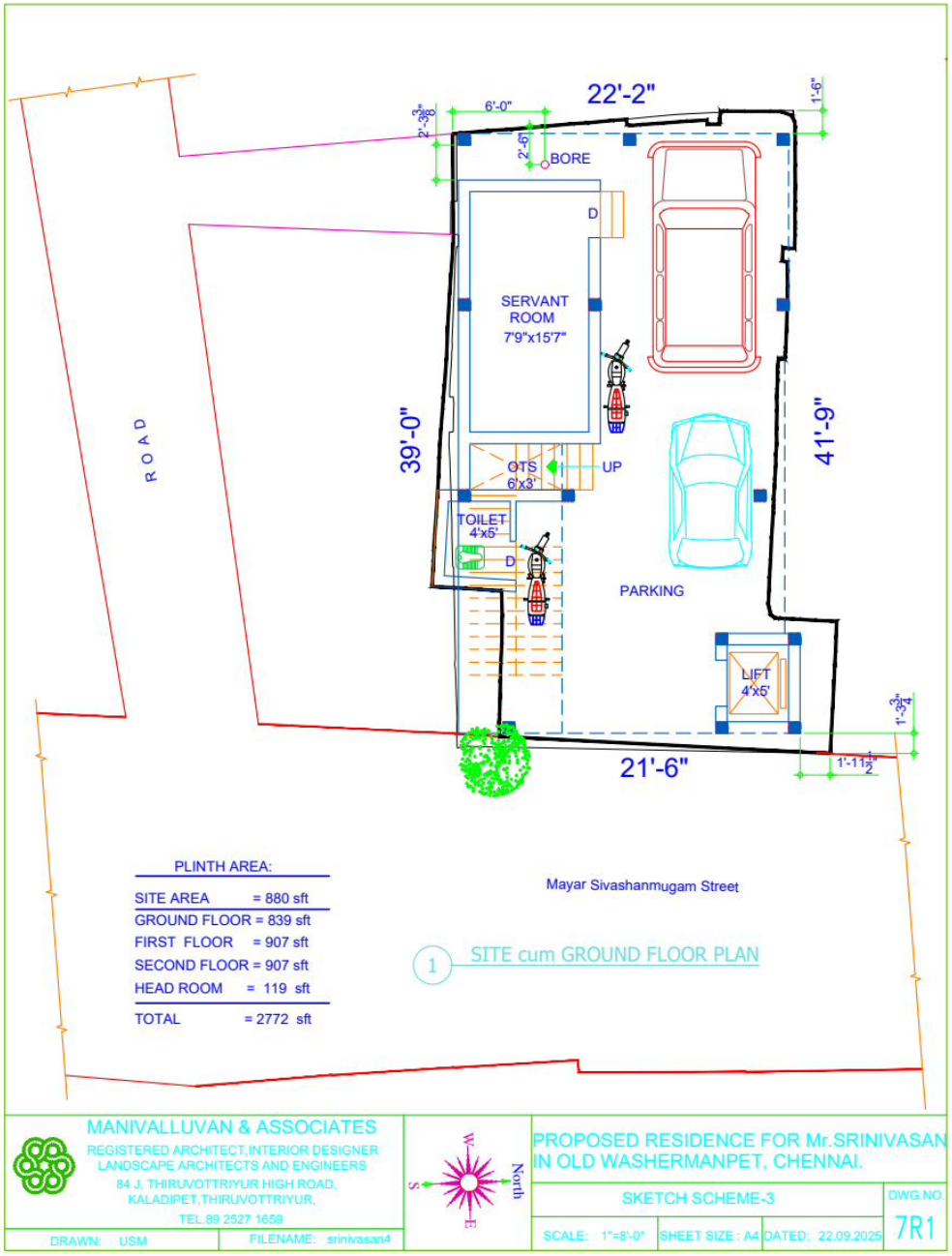
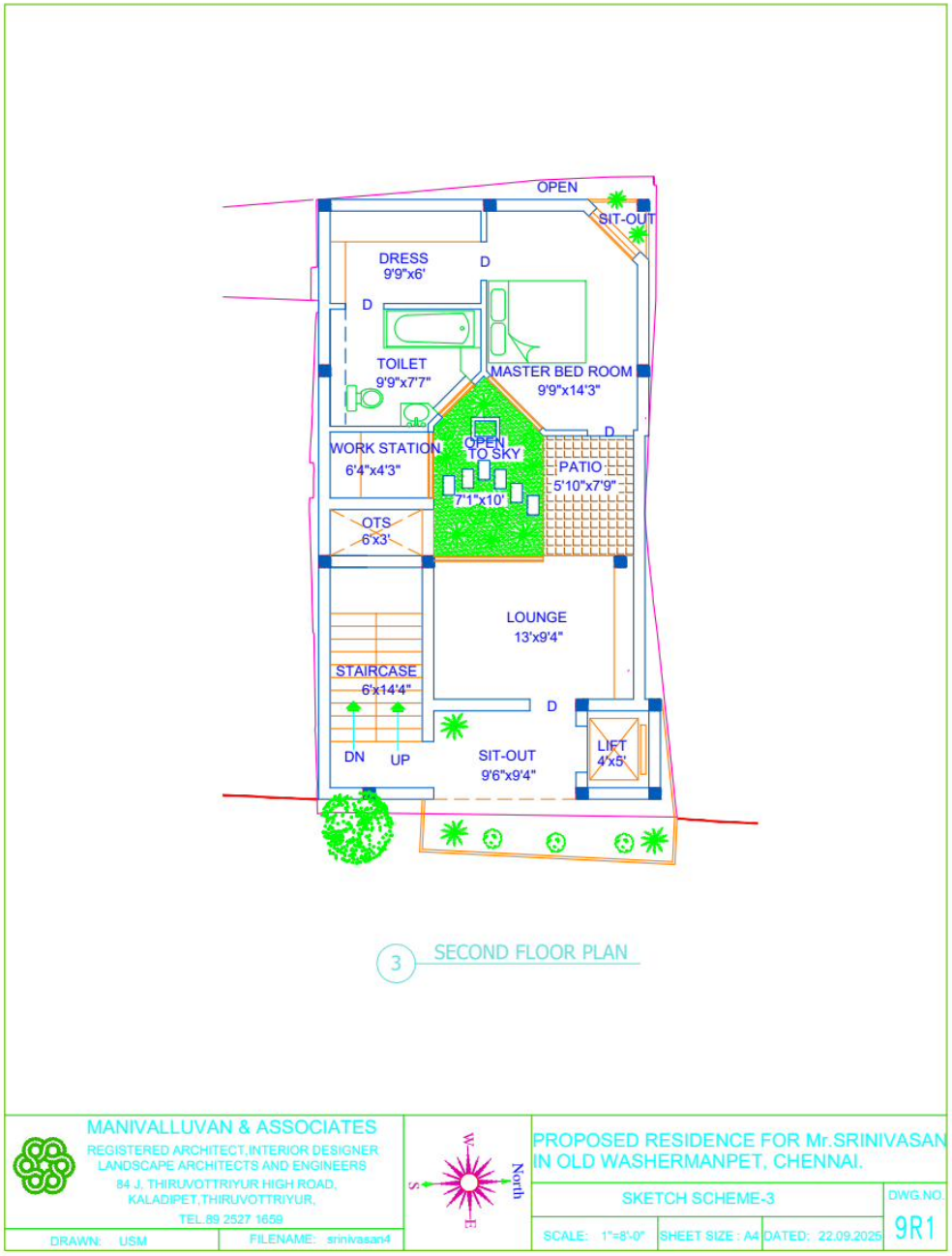
Ground Floor Plan



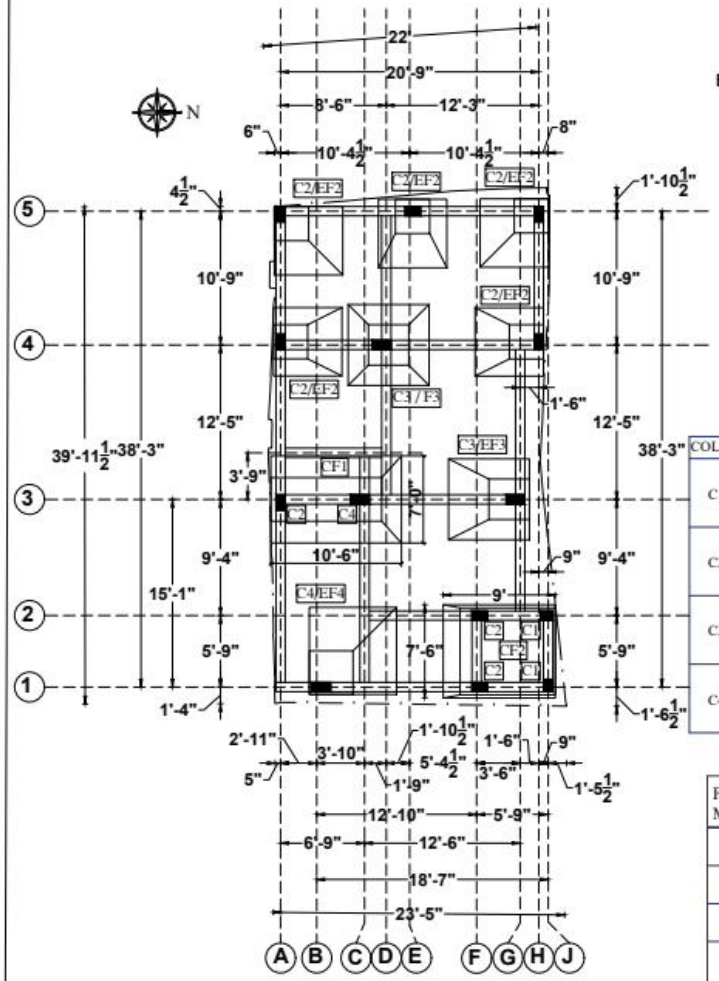
First Floor Plan



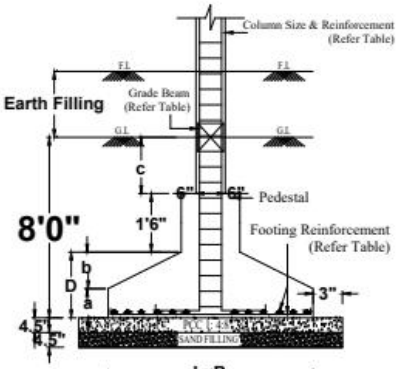
Second Floor Plan



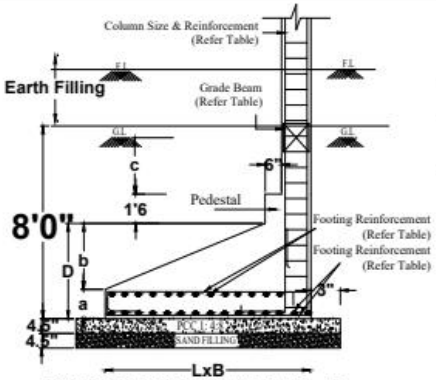
Structural Drawing - Sheet 1



FOUNDATION & PLINTH BEAM LAYOUT



FOOTING DETAILS



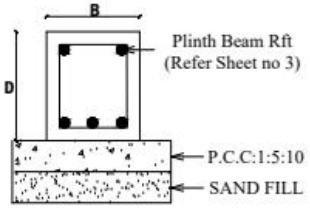
ECCENTRIC FOOTING DETAILS - EF

COLUMN REINFORCEMENT DETAILS

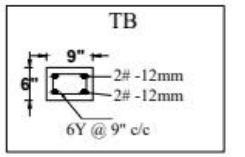
COLUMN MARK	SIZE	NO'S	STILT FLOOR	FIRST FLOOR	SECOND FLOOR	THIRD FLOOR	TERRACE	LIFT ROOM
C1	9" X 1'0"	2	4# -16mm+ 2# -12mm; 8Y @ 6" c/c	4# -16mm+ 2# -12mm; 8Y @ 6" c/c	4# -16mm+ 2# -12mm; 8Y @ 6" c/c	6# -12mm; 8Y @ 6" c/c	4# -12mm; 8Y @ 6" c/c	4# -12mm; 8Y @ 6" c/c
C2	9" X 1'3"	8	6# -16mm+ 2# -12mm; 8Y @ 6" c/c	6# -16mm+ 2# -12mm; 8Y @ 6" c/c	6# -16mm+ 2# -12mm; 8Y @ 6" c/c	8# -12mm; 8Y @ 6" c/c	6# -12mm; 8Y @ 6" c/c	6# -12mm; 8Y @ 6" c/c
C3	9" X 1'6"	2	8# -20mm; 8Y @ 6" c/c	8# -16mm; 8Y @ 6" c/c	8# -16mm; 8Y @ 6" c/c	8# -12mm; 8Y @ 6" c/c	—	—
C4	9" X 1'6"	2	10# -20mm; 8Y @ 6" c/c	10# -16mm; 8Y @ 6" c/c	10# -16mm; 8Y @ 6" c/c	6# -16mm+ 4# -12mm; 8Y @ 6" c/c	8# -12mm; 8Y @ 6" c/c	—

FOOTING REINFORCEMENT DETAILS

FOOTING MARK	NOS	L	B	D	a	b	c	Dia. (mm)	Spacing Main (B)	Spacing Dist (L)	Remarks
EF2	5	5'6"	5'6"	2'9"	1'3"	1'6"	3'9"	12	4"	4"	Single Mat
F3	1	6'6"	6'6"	3'0"	1'3"	1'9"	3'6"	12	6"	6"	Single Mat
EF3	1	6'6"	6'6"	3'0"	1'3"	1'9"	3'6"	12	4"	4"	Single Mat
EF4	1	7'0"	7'0"	3'3"	1'6"	1'9"	3'3"	10	5"	5"	Top Mat
								12	5"	5"	Bottom Mat
CF1	1	10'6"	7'0"	3'0"	1'3"	1'9"	3'6"	12	6"	6"	Top Mat
								12	6"	6"	Bottom Mat
CF2	1	9'0"	7'6"	2'9"	1'3"	1'6"	5'3"	12	6"	6"	Top Mat
								12	6"	6"	Bottom Mat



PLINTH BEAM SECTION DETAILS



ADOPT TIE BEAM DETAIL TB ON TOP LEVEL OF BASEMENT FILLING UNLESS IT EXCEEDS 3'0" IN HEIGHT.

WATER CEMENT RATIO FOR 1 BAG OF CEMENT USE MAXIMUM OF 25 LITRES OF WATER. USE 53 GRADE CEMENT

ALL DIMENSIONS ARE IN FEETS & INCHES

DEPTH OF ALL FOOTING SHOULD BE AT THE SAME LEVEL

LAPPING OF COLUMN RODS SHOULD NOT BE DONE AT THE SAME LEVEL

NO TECHNICAL CHANGES SHOULD BE ADOPTED DUE TO PRACTICAL DIFFICULTIES WITHOUT THE CONFIRMATION OF STRUCTURAL ENGINEER.

NOTES :

1.GRADE OF CONCRETE : M 20 (AS PER IS 456-2000)

2.GRADE OF STEEL Fe 500 (AS PER IS 1786-1985)

3.COVERS.

3.1 FOR GRADE BEAM -1" (25MM)

3.2 FOR FOOTING BOTTOM - 2" (50 MM)

3.3 FOR COLUMN - 1 1/2" (40 MM)

3.4 FOR COLUMN BELOW G.L -2" EXTRA COVER

3.5 FOR FOOTING SIDE - 3" (75 MM)

4. REMOVAL OF FORM WORKS FOR COLUMNS -24 HRS AFTER CASTING.

5. LAP LENGTH FOR ALL BARS SHALL BE 50 x DIA

6.SAFE BEARING CAPACITY OF SOIL IS 16.00 T /m2

7.FOUNDATION IS DESIGNED FOR GROUND+3 UPPER FLOORS ONLY

THIS DRAWING IS THE PROPERTY OF "NJ ASSOCIATES" AND MUST NOT BE PASSED ON TO ANY PERSON OR BODY NOT AUTHORISED BY US TO RECEIVE IT NOR BE COPIED OR OTHERWISE MADE USE EITHER IN FULL OR IN PART BY SUCH PERSON OR BODY WITHOUT OUR PRIOR PERMISSION IN WRITING.

REVISION		
S.NO	DESCRIPTION	APPROVED DATE

PROJECT : Proposed Ground+2 Residential Building at Thiruvotriyur.

CLIENT : Mr. L.Srinivasan.

DRAWING DETAILS : Isolated Foundation Details.

JOB NUMBER: 215/25 | **REVISON NO** - 00

DRAWN BY : J.Santhosh,B.E., | **DRAWING NO**-01/05

SCALE: N.T.S | **DATE :** 09.10.2024

SHEET NUMBER : 01/07 | **SIZE:** A3

STRUCTURAL ENGINEERS:

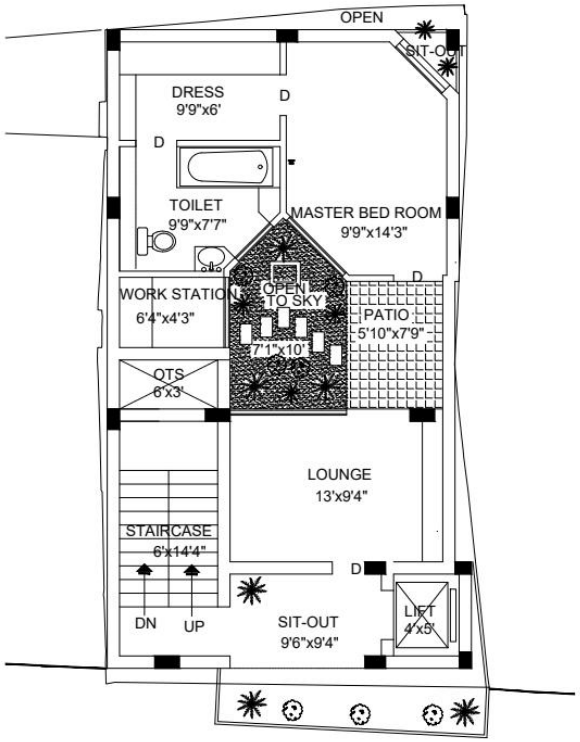
N.JAGADISH,B-Tech.,ME.,

NJ ASSOCIATES

P.No:29, Ganapathy Siva Nagar Main Road,
200ft Ring Road,Opp to Umiya Mahal,
Retteri,Chennai-600 110.
Contact : 90423 79390.
njassociatess@gmail.com

SEAL & SIGNATURE

Structural Drawing - Sheet 2



3 SECOND FLOOR PLAN

MANIVALLUVAN & ASSOCIATES
REGISTERED ARCHITECT,INTERIOR DESIGNER
LANDSCAPE ARCHITECTS AND ENGINEERS
84 J, THIRUVOTTRIYUR HIGH ROAD,
KALADIPET,THIRUVOTTRIYUR,
TEL.89 2527 1659

DRAWN: USM | FILENAME: srinivasan4

W S North E

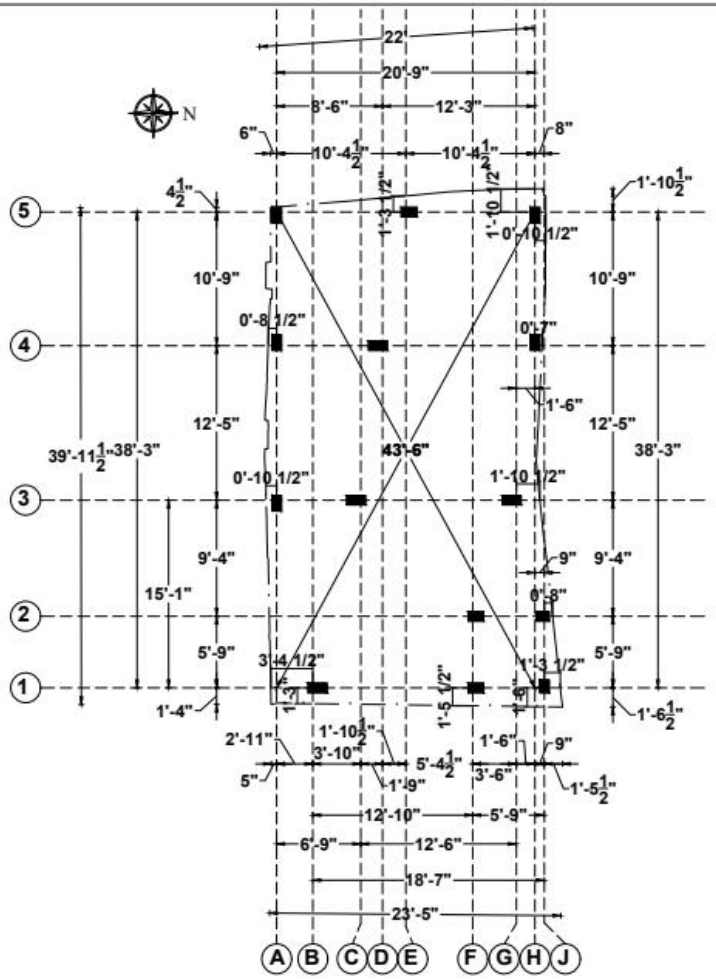
PROPOSED RESIDENCE FOR Mr.SRINIVASAN
IN OLD WASHERMANPET, CHENNAI.

SKETCH SCHEME-3

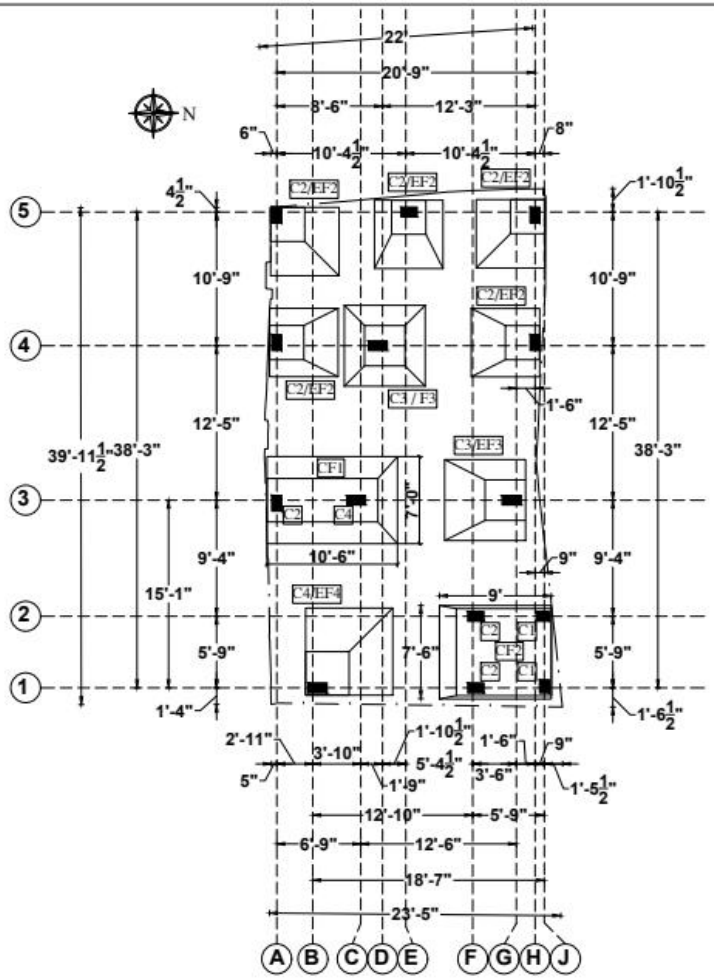
SCALE: 1"=8'-0" | SHEET SIZE : A4 | DATED: 22.09.2025

DWG.NO. 9R1

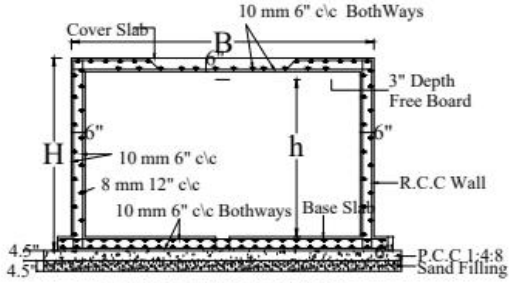
Structural Drawing - Sheet 3



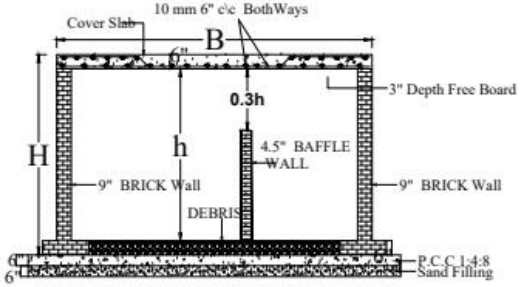
GRID PLAN / CENTRELINE DRAWING



FOUNDATION LAYOUT



SUMP SECTION DETAILS



SEPTIC TANK SECTION DETAILS

WATER CEMENT RATIO FOR 1 BAG OF CEMENT USE MAXIMUM OF 25 LITRES OF WATER.
USE 53 GRADE CEMENT

ALL DIMENSIONS ARE IN FEETS & INCHES

DEPTH OF ALL FOOTING SHOULD BE AT THE SAME LEVEL

LAPPING OF COLUMN RODS SHOULD NOT BE DONE AT THE SAME LEVEL

NO TECHNICAL CHANGES SHOULD BE ADOPTED DUE TO PRACTICAL DIFFICULTIES WITHOUT THE CONFIRMATION OF STRUCTURAL ENGINEER .

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REVISION		
S.NO	DESCRIPTION	APPROVED DATE

PROJECT : Proposed Ground+2 Residential Building at Thiruvotriyur.

CLIENT : Mr. L.Srinivasan.

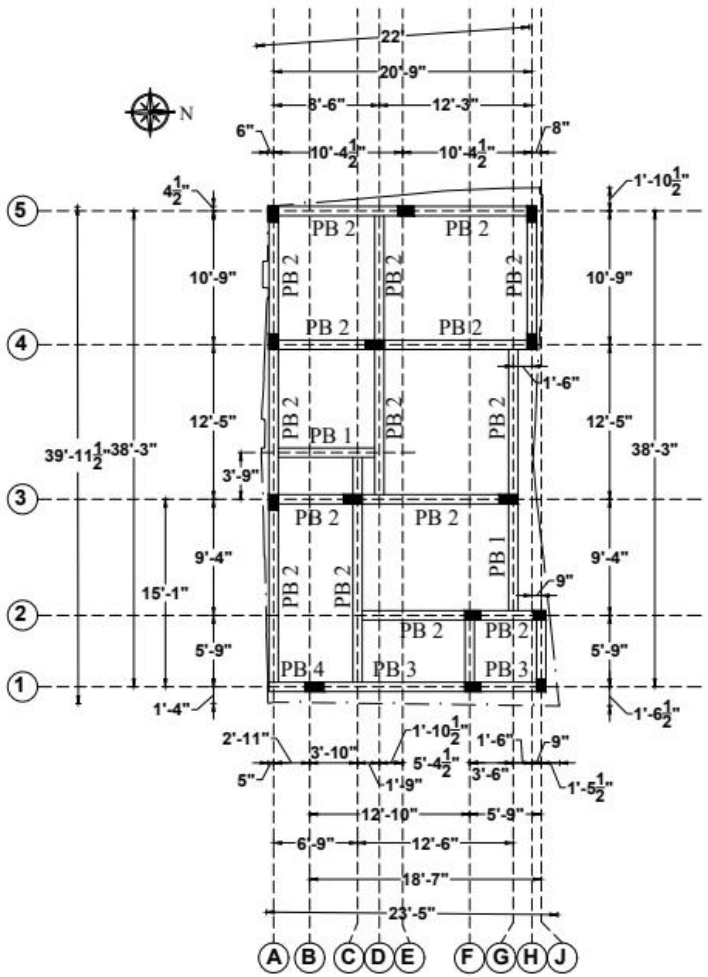
DRAWING DETAILS : Grid Plan , Foundation Layout & UGT Details.

JOB NUMBER: 215/25	REVISON NO - 00
DRAWN BY : J.Santhosh,B.E.,	DRAWING NO-1A/B/05
SCALE: N.T.S	DATE : 09.10.2024
SHEET NUMBER : 02/07	SIZE: A3

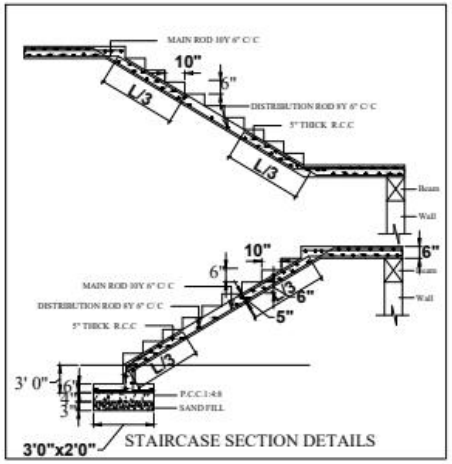
STRUCTURAL ENGINEERS:
N.JAGADISH,B-Tech.,ME.,
NJ ASSOCIATES
P.No:29, Ganapathy Siva Nagar Main Road,
200ft Ring Road,Opp to Umiya Mahal,
Retteri,Chennai-600 110.
Contact : 90423 79390.
njassociatess@gmail.com

SEAL & SIGNATURE

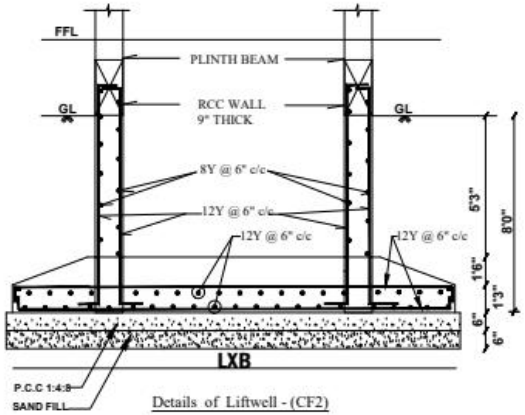
Structural Drawing - Sheet 4



PLINTH BEAM LAYOUT



STAIRCASE SECTION DETAILS



Details of Liftwell - (CF2)

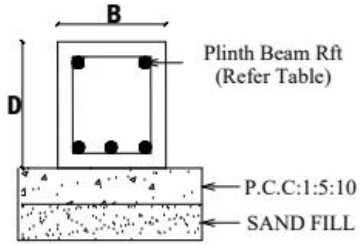
WATER CEMENT RATIO FOR 1 BAG OF CEMENT USE MAXIMUM OF 25 LITRES OF WATER

USE 53 GRADE CEMENT

ALL DIMENSIONS ARE IN FEETS & INCHES

DEPTH OF ALL FOOTING SHOULD BE AT THE SAME LEVEL

LAPPING OF COLUMN RODS SHOULD NOT BE DONE AT SAME LEVEL



PLINTH BEAM
SECTION DETAILS

PLINTH BEAM DETAILS

PB 1	PB 2	PB 3	PB 4
9", 1'3", 2# -12mm, 3# -12mm, 8# @ 6" c/c	9", 1'3", 3# -12mm, 3# -12mm, 8# @ 6" c/c	9", 1'6", 3# -16mm, 3# -16mm, 8# @ 6" c/c	9", 1'6", 3# -16mm, 2# -16mm, 3# -16mm, 8# @ 6" c/c

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REVISION		
S.NO	DESCRIPTION	APPROVED DATE

PROJECT : Proposed Ground+2 Residential Building at Thiruvotriyur.

CLIENT : Mr. L.Srinivasan.

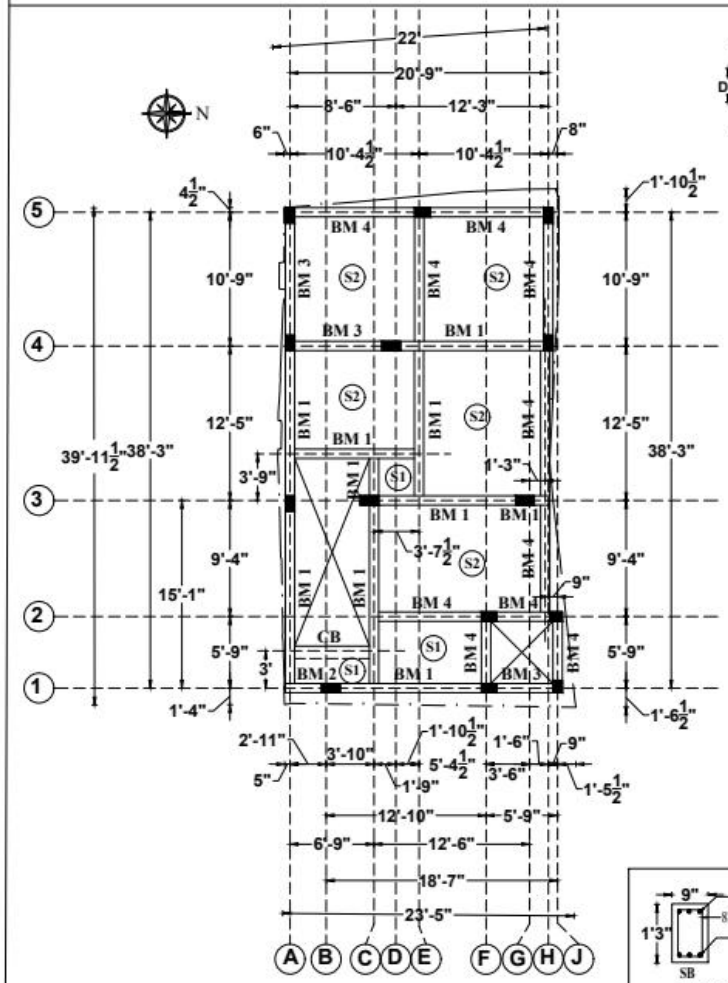
DRAWING DETAILS : Plinth Beam, Staircase & Lift pit Details.

JOB NUMBER: 215/25	REVISON NO - 00
DRAWN BY : J.Santhosh,B.E.,	DRAWING NO-1B/B/05
SCALE: N.T.S	DATE : 09.10.2024
SHEET NUMBER : 03/07	SIZE: A3

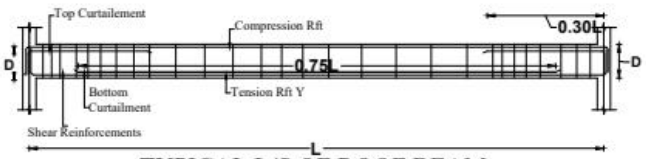
STRUCTURAL ENGINEERS:
N.JAGADISH,B-Tech.,ME.,
NJ ASSOCIATES
P.No:29, Ganapathy Siva Nagar Main Road,
200ft Ring Road,Opp to Umiya Mahal,
Retteri,Chennai-600 110.
Contact : 90423 79390.
njassociatess@gmail.com

SEAL & SIGNATURE

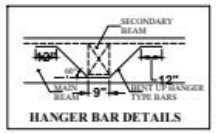
Structural Drawing - Sheet 5



GROUND FLOOR
ROOF BEAM & SLAB LAYOUT

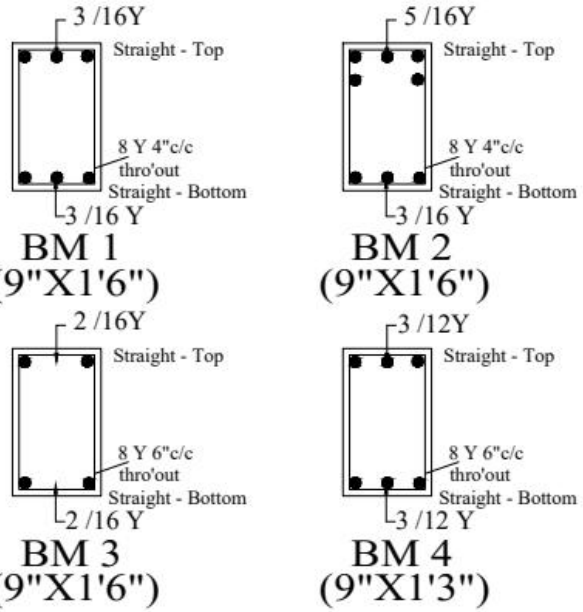


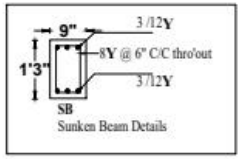
TYPICAL L/S OF ROOF BEAM



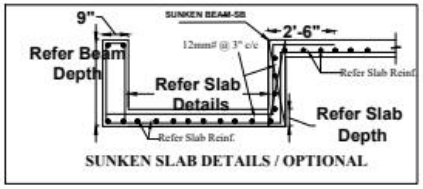
HANGER BAR DETAILS

ADD 2NOS OF 16Y @ POINT LOAD CONDITIONS

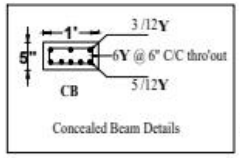




Sunken Beam Details



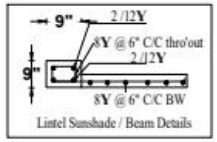
SUNKEN SLAB DETAILS / OPTIONAL



Concealed Beam Details

NOTE:
1.Alternate bars shall be cranked for length of 0.25 L from the support.
2.The cranked bars shall be extented to 0.30 L of adjacent slab.

SLAB NO	SLAB REINFORCEMENT DETAILS				THICKNESS	TYPE
	SHORT SPAN (MAIN)		LONG SPAN (DIS)			
	DIA	SPACING	DIA	SPACING		
S1	10mm	10"	8mm	10"	5"	Alternative Doublemat
S2	10mm	6"	8mm	6"	5"	Twoway



Lintel Sunshade / Beam Details

WATER CEMENT RATIO FOR 1 BAG OF CEMENT USE MAXIMUM OF 25 LITRES OF WATER. USE 53 GRADE CEMENT

ALL DIMENSIONS ARE IN FOOT AND INCHES

BEAM DEPTH IS INCLUSIVE OF SLAB THICKNESS

LAPPING OF COLUMN RODS SHOULD NOT BE DONE AT THE SAME LEVEL.

NOTES :
1.GRADE OF CONCRETE : M 20
(AS PER IS 456-2000)
2.GRADE OF STEEL Fe 500
(AS PER IS 1786-1985)
3. REMOVAL OF FORM WORKS
FOR BEAM SIDES -24 HRS AFTER CASTING.
4. REMOVAL OF FORM WORKS
FOR SLAB BOTTOM -15 DAYS AFTER CASTING.
5. LAP LENGTH FOR ALL BARS SHALL BE 50 x DIA
6.FOUNDATION IS DESIGNED FOR GROUND+3 UPPER FLOORS ONLY

THIS DRAWING IS THE PROPERTY OF "NJ ASSOCIATES" AND MUST NOT BE PASSED ON TO ANY PERSON OR BODY NOT AUTHORISED BY US TO RECEIVE IT NOR BE COPIED OR OTHERWISE MADE USE EITHER IN FULL OR IN PART BY SUCH PERSON OR BODY WITHOUT OUR PRIOR PERMISSION IN WRITING.

REVISION		
S.NO	DESCRIPTION	APPROVED DATE

PROJECT :
Proposed Ground+2 Residential Building at Thiruvotriyur.

CLIENT : Mr. L.Srinivasan.

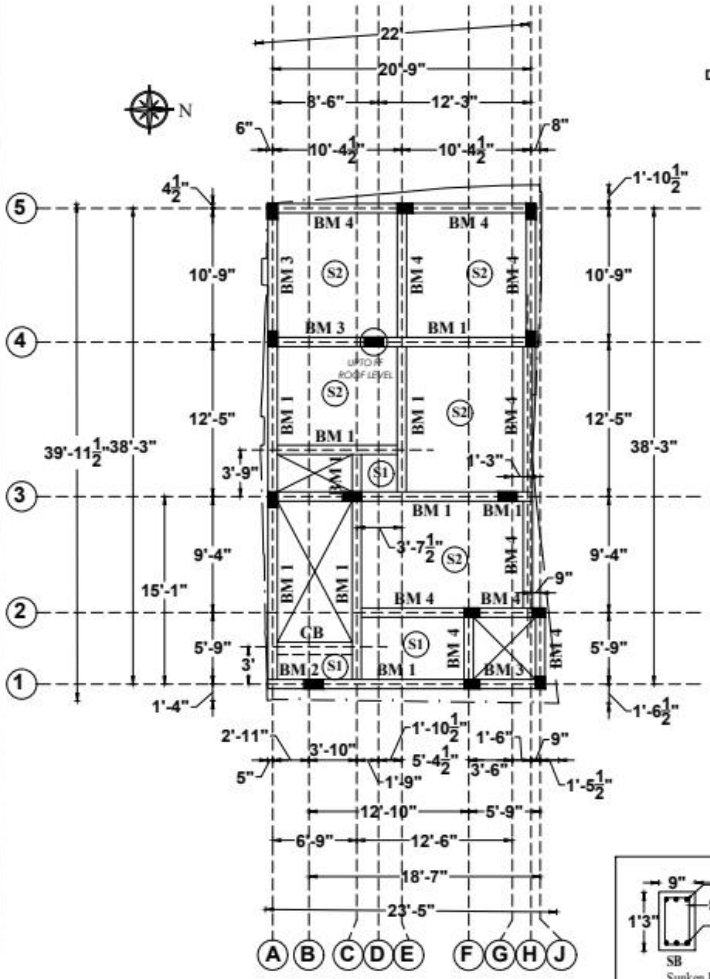
DRAWING DETAILS : Ground Floor Roof Beam & Slab Details.

JOB NUMBER: 215/25 | **REVISON NO** - 00
DRAWN BY : J.Santhosh,B.E., | **DRAWING NO**-02/05
SCALE: N.T.S | **DATE :** 09.10.2024
SHEET NUMBER : 04/07 | **SIZE:** A3

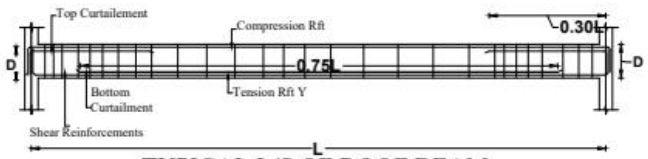
STRUCTURAL ENGINEERS:
N.JAGADISH,B-Tech.,ME.,
NJ ASSOCIATES
P.No:29, Ganapathy Siva Nagar Main Road,
200ft Ring Road,Opp to Umiya Mahal,
Retteri,Chennai-600 110.
Contact : 90423 79390.
njassociatess@gmail.com

SEAL & SIGNATURE

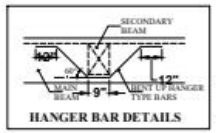
Structural Drawing - Sheet 6



FIRST FLOOR
ROOF BEAM & SLAB LAYOUT

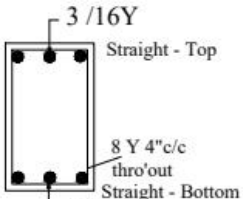


TYPICAL L/S OF ROOF BEAM

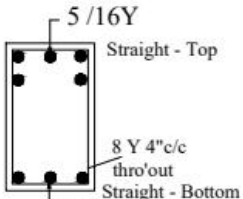


HANGER BAR DETAILS

ADD 2NOS OF 16Y @ POINT LOAD CONDITIONS



BM 1
(9"X1'6")



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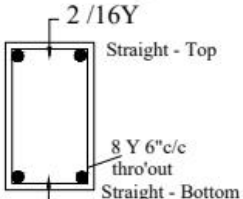
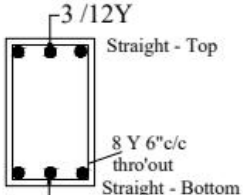
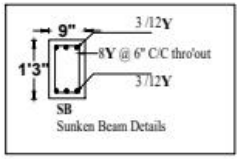
(9"X1'6")



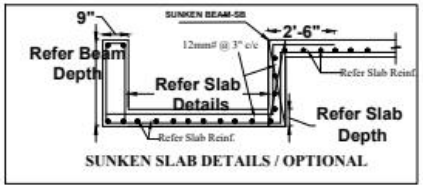
BM 3
(9"X1'6")



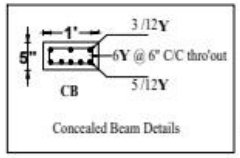
BM 4
(9"X1'3")



Sunken Beam Details



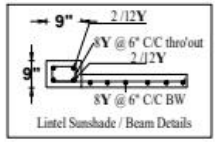
SUNKEN SLAB DETAILS / OPTIONAL



Concealed Beam Details

NOTE:
1.Alternate bars shall be cranked for length of 0.25 L from the support.
2.The cranked bars shall be extented to 0.30 L of adjacent slab.

SLAB NO	SLAB REINFORCEMENT DETAILS				THICKNESS	TYPE
	SHORT SPAN (MAIN)		LONG SPAN (DIS)			
	DIA	SPACING	DIA	SPACING		
S1	10mm	10"	8mm	10"	5"	Alternative Doublemat
S2	10mm	6"	8mm	6"	5"	Twoway



Lintel Sunshade / Beam Details

WATER CEMENT RATIO FOR 1 BAG OF CEMENT USE MAXIMUM OF 25 LITRES OF WATER. USE 53 GRADE CEMENT

ALL DIMENSIONS ARE IN FOOT AND INCHES

BEAM DEPTH IS INCLUSIVE OF SLAB THICKNESS

LAPPING OF COLUMN RODS SHOULD NOT BE DONE AT THE SAME LEVEL.

NOTES :
1.GRADE OF CONCRETE : M 20
(AS PER IS 456-2000)
2.GRADE OF STEEL Fe 500
(AS PER IS 1786-1985)
3. REMOVAL OF FORM WORKS
FOR BEAM SIDES -24 HRS AFTER CASTING.
4. REMOVAL OF FORM WORKS
FOR SLAB BOTTOM -15 DAYS AFTER CASTING.
5. LAP LENGTH FOR ALL BARS SHALL BE 50 x DIA
6.FOUNDATION IS DESIGNED FOR GROUND+3 UPPER FLOORS ONLY

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REVISION		
S.NO	DESCRIPTION	APPROVED DATE

PROJECT :
Proposed Ground+2 Residential Building at Thiruvotriyur.

CLIENT : Mr. L.Srinivasan.

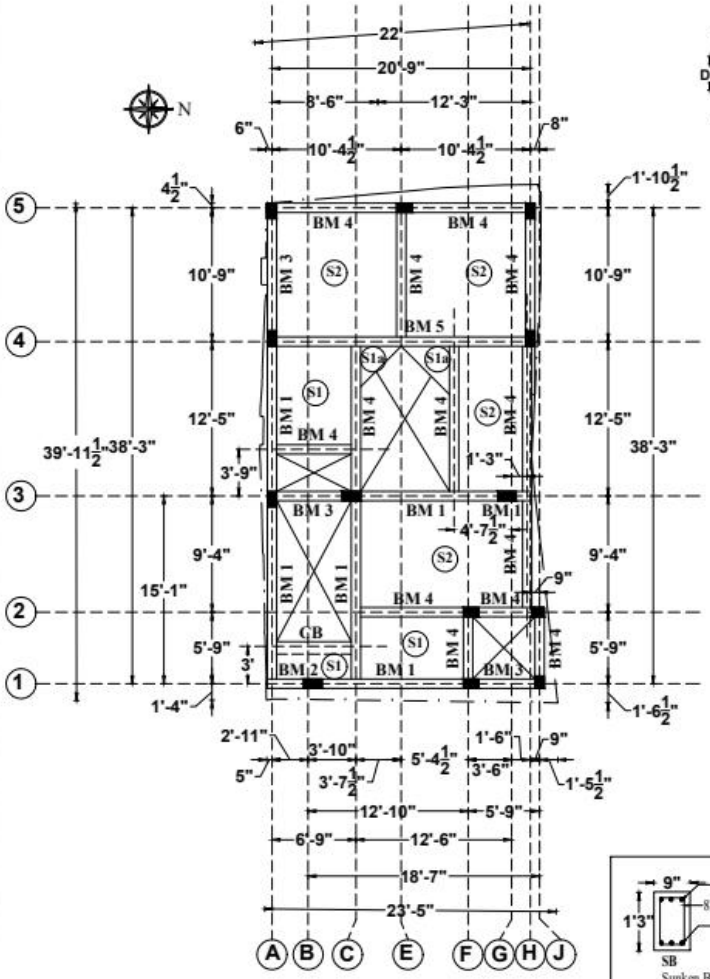
DRAWING DETAILS : First Floor Roof Beam & Slab Details.

JOB NUMBER: 215/25 | **REVISON NO** - 00
DRAWN BY : J.Santhosh,B.E., | **DRAWING NO**-03/05
SCALE: N.T.S | **DATE :** 09.10.2024
SHEET NUMBER : 05/07 | **SIZE:** A3

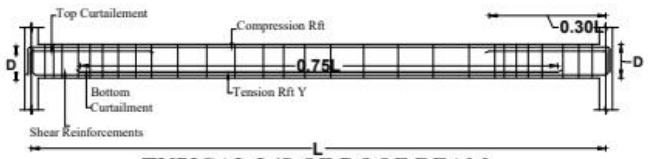
STRUCTURAL ENGINEERS:
N.JAGADISH,B-Tech.,ME.,
NJ ASSOCIATES
P.No:29, Ganapathy Siva Nagar Main Road,
200ft Ring Road,Opp to Umiya Mahal,
Retteri,Chennai-600 110.
Contact : 90423 79390.
njassociatess@gmail.com

SEAL & SIGNATURE

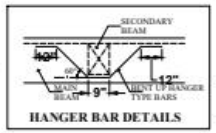
Structural Drawing - Sheet 7



SECOND FLOOR
ROOF BEAM & SLAB LAYOUT

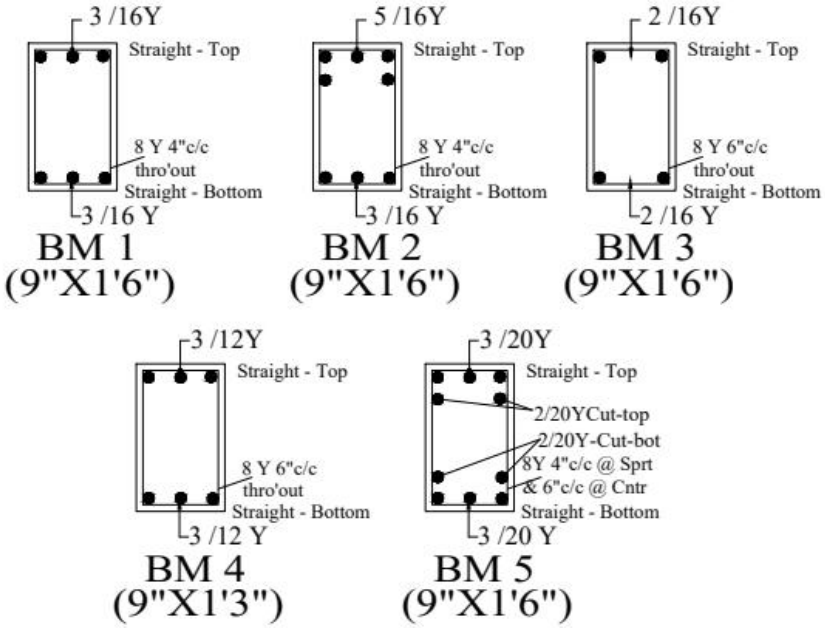


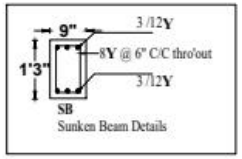
TYPICAL L/S OF ROOF BEAM



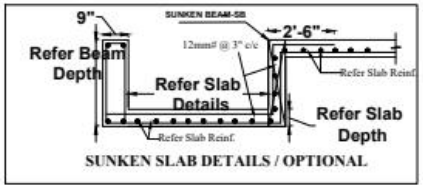
HANGER BAR DETAILS

ADD 2NOS OF 16Y @ POINT LOAD CONDITIONS

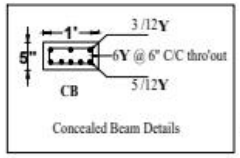




Sunken Beam Details



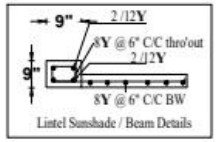
SUNKEN SLAB DETAILS / OPTIONAL



Concealed Beam Details

NOTE:
1.Alternate bars shall be cranked for length of 0.25 L from the support.
2.The cranked bars shall be extented to 0.30 L of adjacent slab.

SLAB NO	SLAB REINFORCEMENT DETAILS				THICKNESS	TYPE
	SHORT SPAN (MAIN)		LONG SPAN (DIS)			
	DIA	SPACING	DIA	SPACING		
S1a	10mm	6"	8mm	6"	5"	Cantilever
S1	10mm	10"	8mm	10"	5"	Alternative Doublemat
S2	10mm	6"	8mm	6"	5"	Twoway



Lintel Sunshade / Beam Details

WATER CEMENT RATIO FOR 1 BAG OF CEMENT USE MAXIMUM OF 25 LITRES OF WATER. USE 53 GRADE CEMENT

ALL DIMENSIONS ARE IN FOOT AND INCHES

BEAM DEPTH IS INCLUSIVE OF SLAB THICKNESS

LAPPING OF COLUMN RODS SHOULD NOT BE DONE AT THE SAME LEVEL.

NOTES :
1.GRADE OF CONCRETE : M 20
(AS PER IS 456-2000)
2.GRADE OF STEEL Fe 500
(AS PER IS 1786-1985)
3. REMOVAL OF FORM WORKS
FOR BEAM SIDES -24 HRS AFTER CASTING.
4. REMOVAL OF FORM WORKS
FOR SLAB BOTTOM -15 DAYS AFTER CASTING.
5. LAP LENGTH FOR ALL BARS SHALL BE 50 x DIA
6.FOUNDATION IS DESIGNED FOR GROUND+3
UPPER FLOORS ONLY

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REVISION		
S.NO	DESCRIPTION	APPROVED DATE

PROJECT :
Proposed Ground+2 Residential Building at Thiruvotriyur.

CLIENT : Mr. L.Srinivasan.

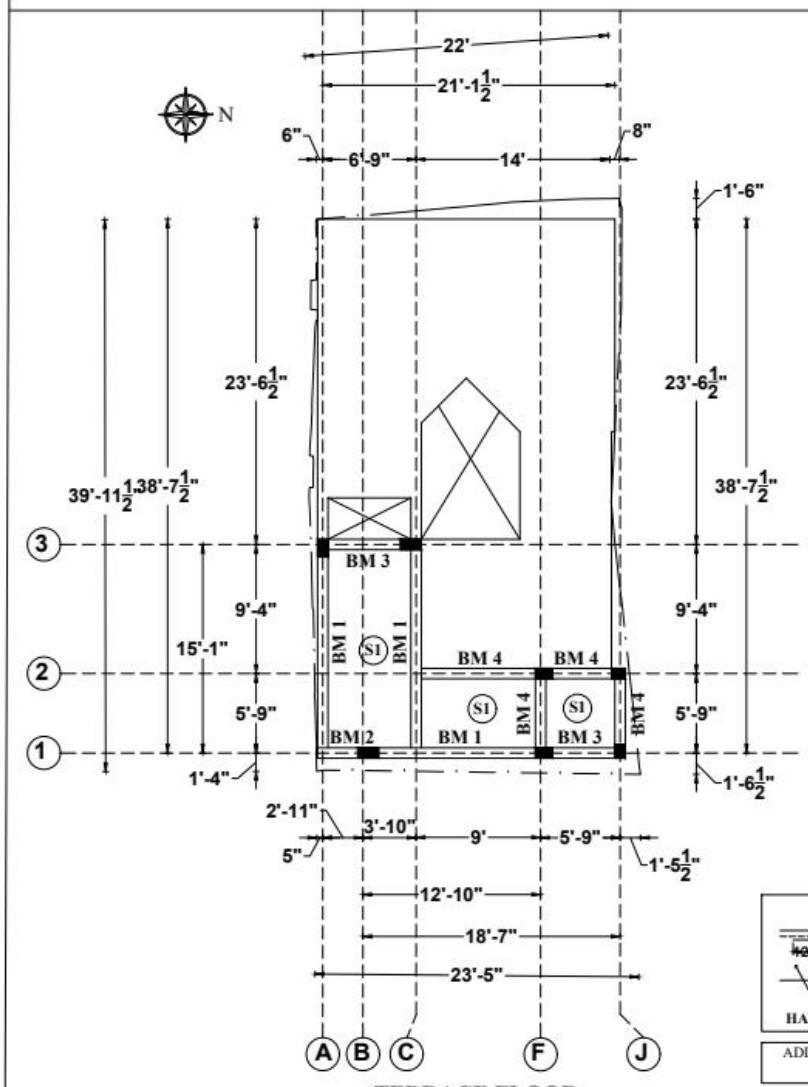
DRAWING DETAILS : Second Floor Roof Beam & Slab Details.

JOB NUMBER: 215/25 | **REVISON NO** - 00
DRAWN BY : J.Santhosh,B.E., | **DRAWING NO**-04/05
SCALE: N.T.S | **DATE :** 09.10.2024
SHEET NUMBER : 06/07 | **SIZE:** A3

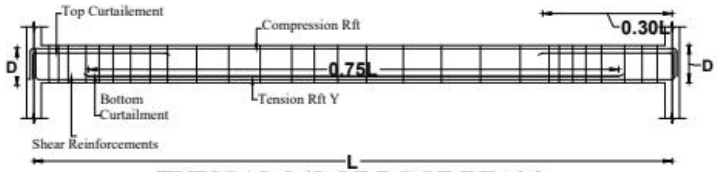
STRUCTURAL ENGINEERS:
N.JAGADISH,B-Tech.,ME.,
NJ ASSOCIATES
P.No:29, Ganapathy Siva Nagar Main Road,
200ft Ring Road,Opp to Umiya Mahal,
Retteri,Chennai-600 110.
Contact : 90423 79390.
njassociatess@gmail.com

SEAL & SIGNATURE

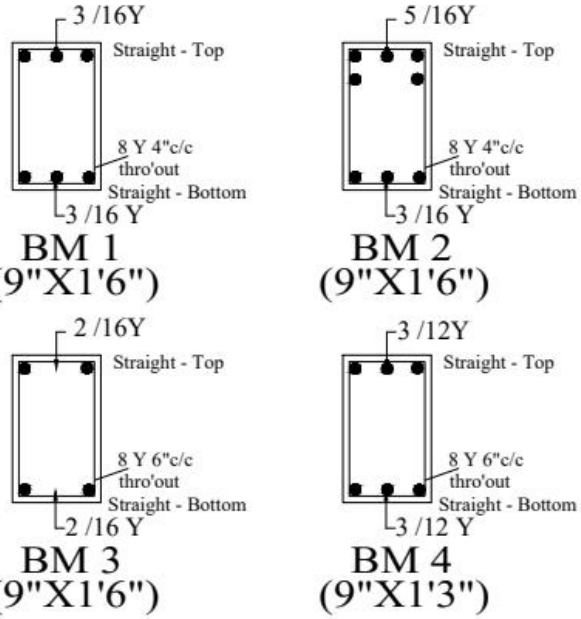
Structural Drawing - Sheet 8

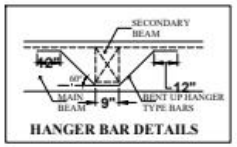


TERRACE FLOOR
ROOF BEAM & SLAB LAYOUT



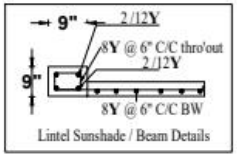
TYPICAL L/S OF ROOF BEAM





HANGER BAR DETAILS

ADD 2NOS OF 16Y @ POINT LOAD CONDITIONS



Lintel Sunshade / Beam Details

WATER CEMENT RATIO FOR 1 BAG OF CEMENT USE MAXIMUM OF 25 LITRES OF WATER.
USE 53 GRADE CEMENT

ALL DIMENSIONS ARE IN FOOT AND INCHES

BEAM DEPTH IS INCLUSIVE OF SLAB THICKNESS

LAPPING OF COLUMN RODS SHOULD NOT BE DONE AT THE SAME LEVEL.

NOTE:
1. Alternate bars shall be cranked for length of 0.25 L from the support.
2. The cranked bars shall be extented to 0.30 L of adjacent slab.

SLAB NO	SLAB REINFORCEMENT DETAILS				THICKNESS	TYPE
	SHORT SPAN (MAIN)		LONG SPAN (DIS)			
	DIA	SPACING	DIA	SPACING		
S1	10mm	6"	8mm	6"	5"	Oneway

NOTES :
1. GRADE OF CONCRETE : M 20
(AS PER IS 456-2000)
2. GRADE OF STEEL Fe 500
(AS PER IS 1786-1985)
3. REMOVAL OF FORM WORKS
FOR BEAM SIDES -24 HRS AFTER CASTING.
4. REMOVAL OF FORM WORKS
FOR SLAB BOTTOM -15 DAYS AFTER CASTING.
5. LAP LENGTH FOR ALL BARS SHALL BE 50 x DIA
6. FOUNDATION IS DESIGNED FOR GROUND+3 UPPER FLOORS ONLY

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REVISION		
S.NO	DESCRIPTION	APPROVED DATE

PROJECT :
Proposed Ground+2 Residential Building at Thiruvotriyur.

CLIENT : Mr. L.Srinivasan.

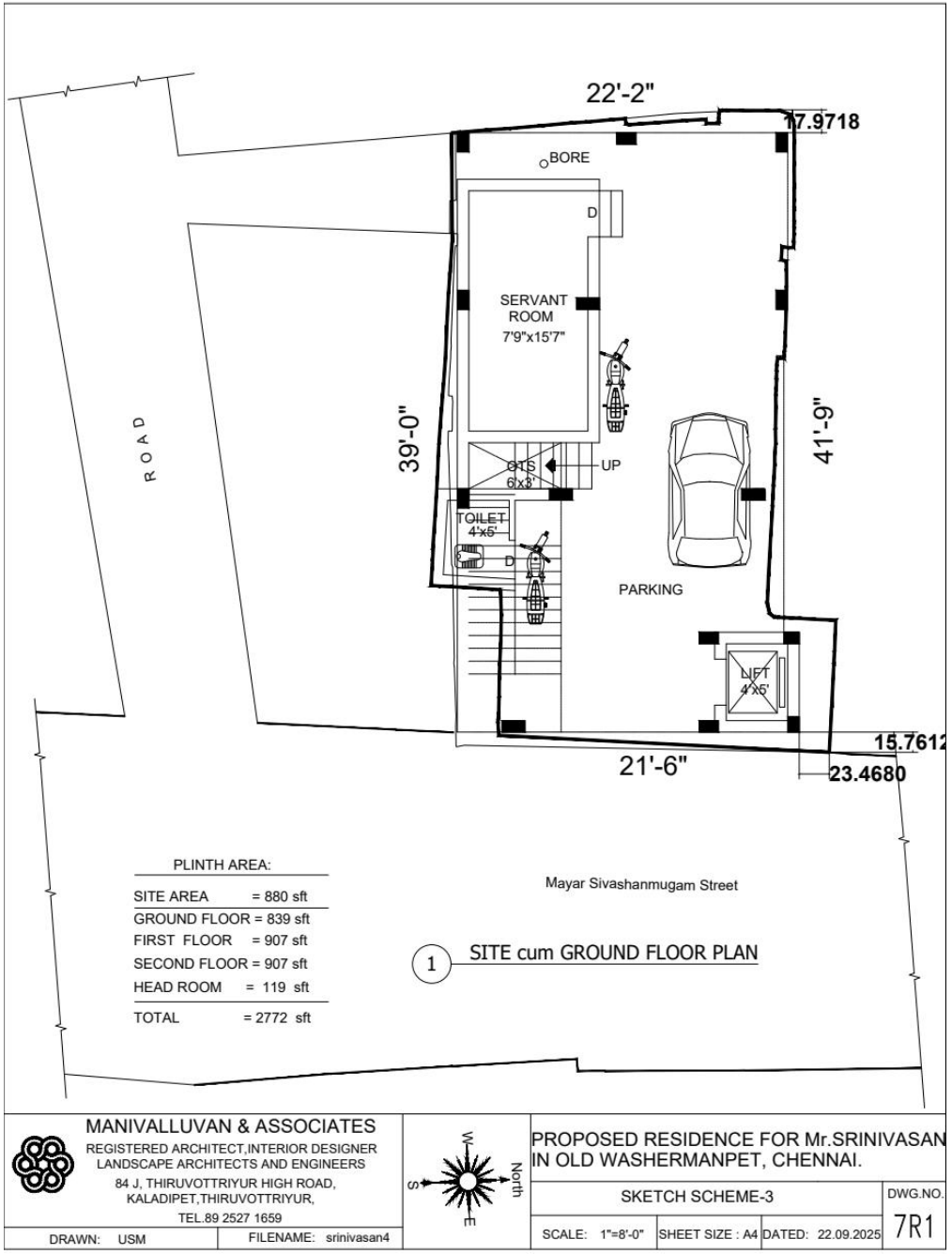
DRAWING DETAILS : Terrace Floor
Roof Beam & Slab Details.

JOB NUMBER: 215/25 | **REVISON NO** - 00
DRAWN BY : J.Santhosh,B.E., | **DRAWING NO**-05/05
SCALE: N.T.S | **DATE :** 09.10.2024
SHEET NUMBER : 07/07 | **SIZE:** A3

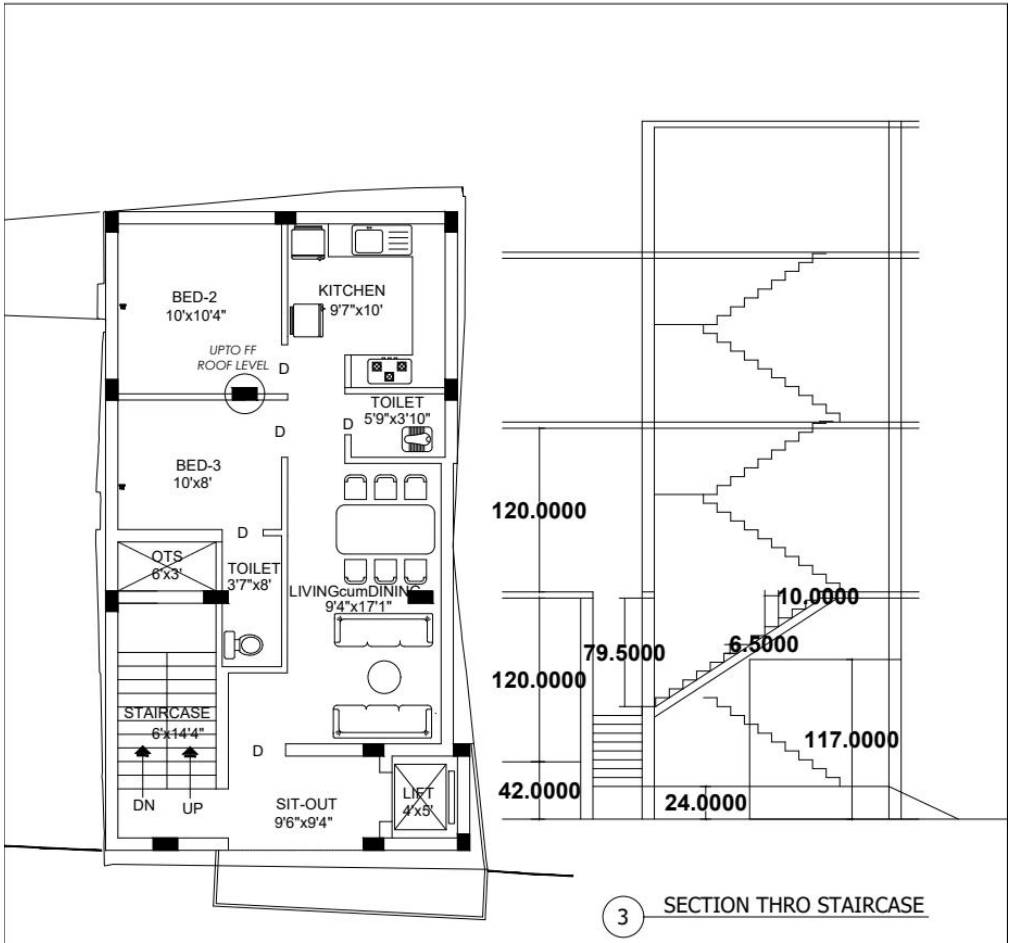
STRUCTURAL ENGINEERS:
N.JAGADISH,B-Tech.,ME.,
NJ ASSOCIATES
P.No:29, Ganapathy Siva Nagar Main Road,
200ft Ring Road,Opp to Umiya Mahal,
Retteri,Chennai-600 110.
Contact : 90423 79390.
njassociatess@gmail.com

SEAL & SIGNATURE

Structural Drawing - Sheet 9



Structural Drawing - Sheet 10



MANIVALLUVAN & ASSOCIATES
REGISTERED ARCHITECT,INTERIOR DESIGNER
LANDSCAPE ARCHITECTS AND ENGINEERS
84 J, THIRUVOTTRIYUR HIGH ROAD,
KALADIPET,THIRUVOTTRIYUR,
TEL.89 2527 1659

DRAWN: USM | FILENAME: srinivasan4

W S North E

PROPOSED RESIDENCE FOR Mr.SRINIVASAN
IN OLD WASHERMANPET, CHENNAI.

SKETCH SCHEME-3

SCALE: 1"=8'-0" | SHEET SIZE : A4 | DATED: 22.09.2025 | DWG.NO. 8R1